

# Technology Mediated Responding

SurvMeth 632  
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# Outline

## Impact of computing device on I-R interaction

- Intermediation, division of attention

## Self-administered questionnaires

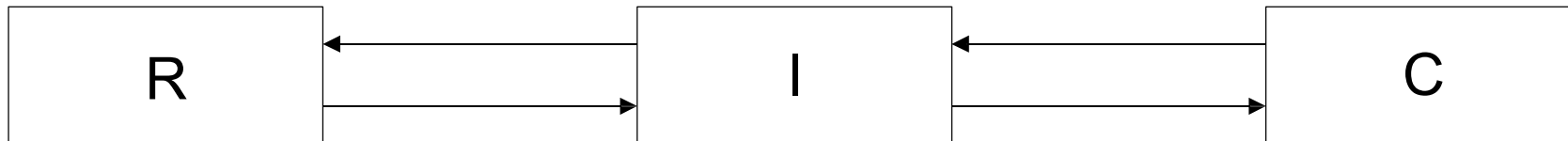
- Sensitive questions; social presence; embodiment
- Non-sensitive questions: requests for clarification

## Web Surveys

- Interactive features; effort; initiative

# Human versus Computer Mediation

- In interviewer-administered questionnaires, interviewer (I) mediates between computer (C) and respondent (R): I enters info on behalf of R
  - Human-Human-Computer interaction
  - Similar to interactions between doctor and patient, airline reservationist and client, etc.



- In social media, help chats, email, etc., the computer mediates between users (Computer Mediated Communication or “CMC”)



R = respondent, I = interviewer, U = user, C = computer

# Intermediation

## Human-human-computer interaction

- Interviewer (*Iwer*) is *surrogate user* or *intermediary*
- Respondent (*R*) is *end-user*
  - *R* interacts with computer (*C*) through *Iwer* who enters responses
- *Iwer* must split attention between *R* and *C*
  - If *R* wants to change answer, *Iwer* must back up then jump forward
  - If *R* provides answer in format not accepted by *C*, *Iwer* must translate, e.g., re-format a date or convert units
  - If *Iwer* has trouble using *C*, i.e., a *usability problem*, more attention must be devoted to *C* and less to *R*
- In general, if *Iwer* is paying attention to *C*, they are not attending to *R* which makes it hard to monitor *R*'s comprehension or ability to answer the question

# Intermediation (2)

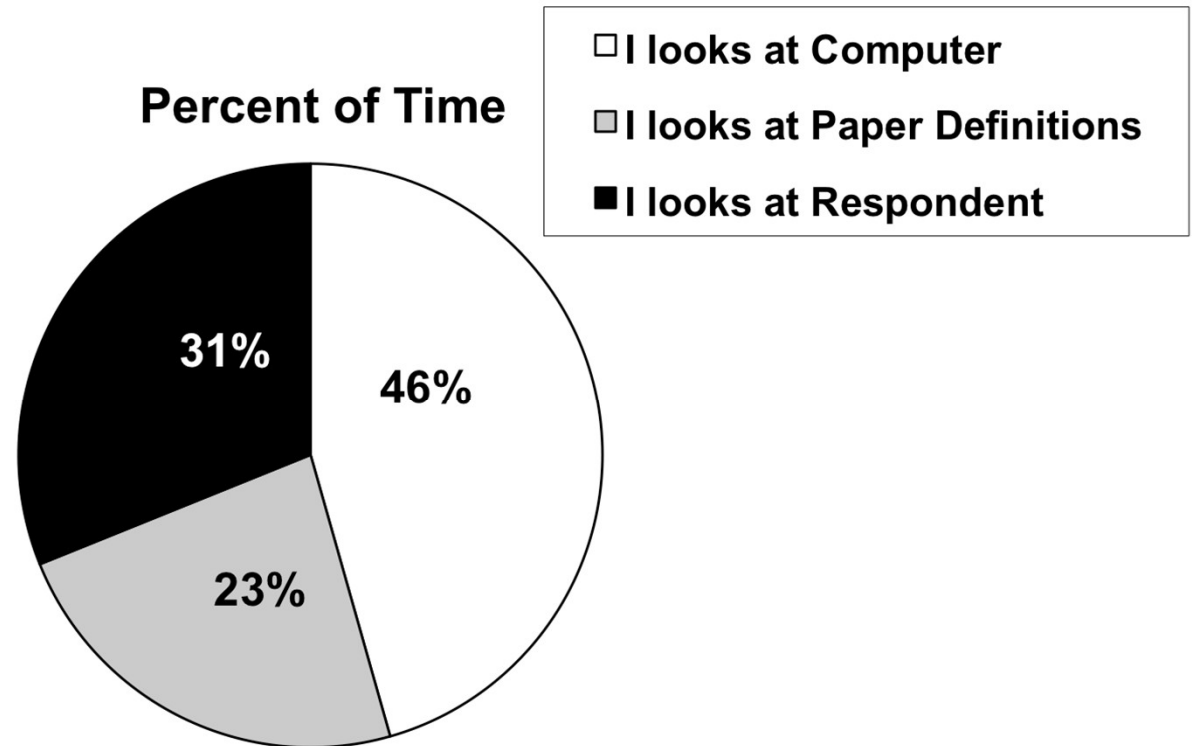
- Directory Assistance operators as intermediaries (Lawrence, Atwood, Dews, 1994)
  - Like *Iwer*, Operators mediate between context of everyday life and technical discourse, e.g., translate requests into search strings likely to succeed:
    - omit words (e.g., “diner”)
    - expand customer’s abbreviations (“assoc” for “AHRC”)
    - abbreviate customer request (“C-E-N-T H-A-R-D” for “central hardware”)
    - enter as stated “KMART” for “K-Mart”
  - Fast operators are better at typing search string while customer speaking

# Mediating between social and technical interaction

|                                                                                   | Faster Operator | Slower Operators |
|-----------------------------------------------------------------------------------|-----------------|------------------|
| % of keying during customer's turns                                               | 38.18           | 34.24            |
| % keying during (all) pauses                                                      | 29.52           | 38.4             |
| Mean length (msec) of interspeaker pauses (customer and operator pauses combined) | 362             | 454              |
| Intraspeaker pauses as % of turns                                                 | 9.94            | 12.88            |

Lawrence et al. (1994)

# Interviewers divide visual attention between *R*, *C*, and other materials



Conrad, Schober & Dijkstra, 2004\*

\*in Q6

# Computerized Self Administration

- Human mediation (e.g., *Iwer* asks questions and records answers) is likely to introduce “social presence” – sense that someone else is there, even if on phone
- Removing *Iwer* seems to eliminate social presence and increase privacy
  - Should reduce *Rs*’ self-presentation concerns and increase disclosure of sensitive info
- Self-administered (paper) questionnaires long used to promote disclosure of sensitive information in interviews
- Computer Assisted Self Interviewing (CASI) extends idea to computerized administration
  - *Iwer* gives laptop/tablet to *R* to self-administer sensitive Qs
  - Textual Qs displayed on laptop/tablet and *R* directly enters answers
- Audio (ACASI) presentation of Qs assumed to further increase privacy
- More socially undesirable responses in self-administered than *Iwer*-administered mode considered evidence of greater candor
  - No incentive to falsely provide undesirable answers but external validation is rare

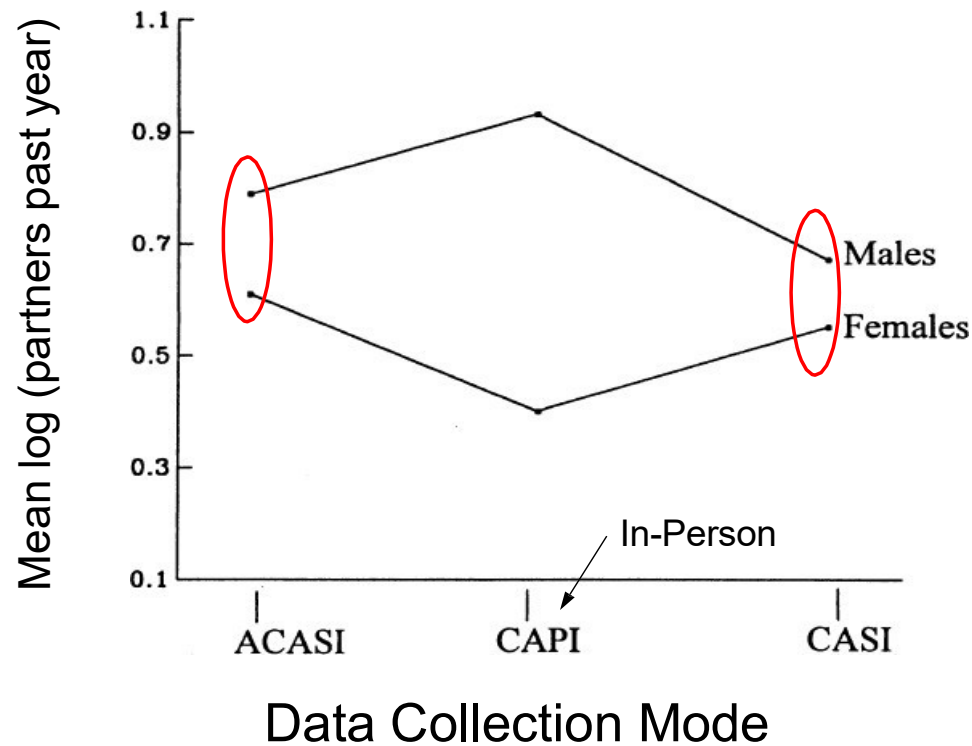


# CASI, ACASI and Disclosure

- Tourangeau & Smith (1996) compare data collection with and without interviewer, with and without computer
- In general, men reported 2-4 times as many heterosexual sex partners as women; presumed due to socially desirable reporting in opposite directions for men and women
- Self-administration reduced (apparent) socially desirable reporting
  - in “Women’s Health Study,” CASI led to more reported partners in past year and past 5 years than did FTF or SAQ (paper self-admin)
  - In “ACASI study,” asymmetry between sexes for # reported heterosexual sex partners is smaller for self-administration (CASI and ACASI) than for In-Person interviews (CAPI)

# Reported sex partners by gender and mode

(Tourangeau & Smith, 1996)



# CASI, ACASI and Disclosure (2)

- Tourangeau & Smith (1996) (cont'd)
  - More illicit drug use reported with self-administration (ACASI and CASI) than in interview (CAPI)

|           | Time Frame |           |          |
|-----------|------------|-----------|----------|
|           | Past Month | Past Year | Lifetime |
| Cocaine   | 1.74       | 2.84      | 1.81     |
| Marijuana | 1.66       | 1.61      | 1.48     |

Ratio of Disclosure in Self-administration to Interviewer-administration

- Extra benefit from audio relative to other forms of self-administration: # sex partners reported overall ACASI > CASI > CAPI

# Why do *Rs* disclose more in ACASI than FTF?

- Absence of human interviewer? Absence of human-like features such as a face?
- Can test by comparing disclosure in ACASI and In-Person (extremes) to disclosure with Virtual Interviewers (VIs), somewhere in between the extremes
  - clearly automated but embodied
- Lind, Schober, Conrad & Reichert (2013) compared prevalence of undesirable answers in 4 modes:
  - (1) FTF (In-person), (2) High Animation (HA) VI, (3) Low Animation (LA) VI, (4) ACASI
  - T&S ('96) findings would suggest automated modes (ACASI, HA and LA) will promote disclosure relative to FTF interviews
  - But if virtual interviewers create *social presence* could inhibit disclosure, producing results more like FTF

# ACASI vs. Interviewer (CAPI)

| Dimension             | Human interviewer                                                                                                                                                                                                                                                                                                                     | ACASI                                                                                                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dialogue capability   | <p>Interviewer produces scripted utterances in real time in ways that will never be exactly the same for all respondents.</p> <p>Interviewers can, potentially, produce (scripted or unscripted) neutral probes or even clarification as needed.</p>                                                                                  | <p>ACASI speech is prerecorded and each question is delivered in precisely the same way every time it is administered.</p> <p>ACASI systems do not generally produce probes or clarification (although in principle they could).</p>                                                                                              |
| Embodiment            | <p>Interviewer's body and face can move and express her cognitive and affective states while speaking as while listening to answers and entering them into the computer.</p> <p>By their very presence interviewers display both visual and auditory cues of their demographic characteristics—their age, ethnicity, gender, etc.</p> | <p>ACASI systems have no bodily or facial display (see Cassell &amp; Miller, 2008, p. 165-166, Foucault, 2009, and von der Pütten et al., 2011, for discussion of potential displays by virtual interviewing systems of the future).</p> <p>ACASI systems only have auditory cues of the recorded interviewer's demographics.</p> |
| Perceptual capability | <p>Interviewer can observe the <i>respondent's</i> physical characteristics and displays, and can make inferences from them (valid or not) about the respondent's cognitive and affective states, the physical condition and circumstances, and demographic characteristics that might be relevant to interpreting responses.</p>     | <p>ACASI system cannot observe or make inferences about the respondent (though in principle more advanced systems could; see Picard, 1998).</p>                                                                                                                                                                                   |
| Evaluative capability | <p>Interviewer has the ability to judge (positively or negatively) the sensitive behaviors that respondents report in the interview.</p>                                                                                                                                                                                              | <p>The ACASI system cannot judge.</p>                                                                                                                                                                                                                                                                                             |

Table 1. Lind, Schober, Conrad, Reichert (2013)

# Virtual Interviewer Interface

Lind, Schober, Conrad & Reichert (2013)



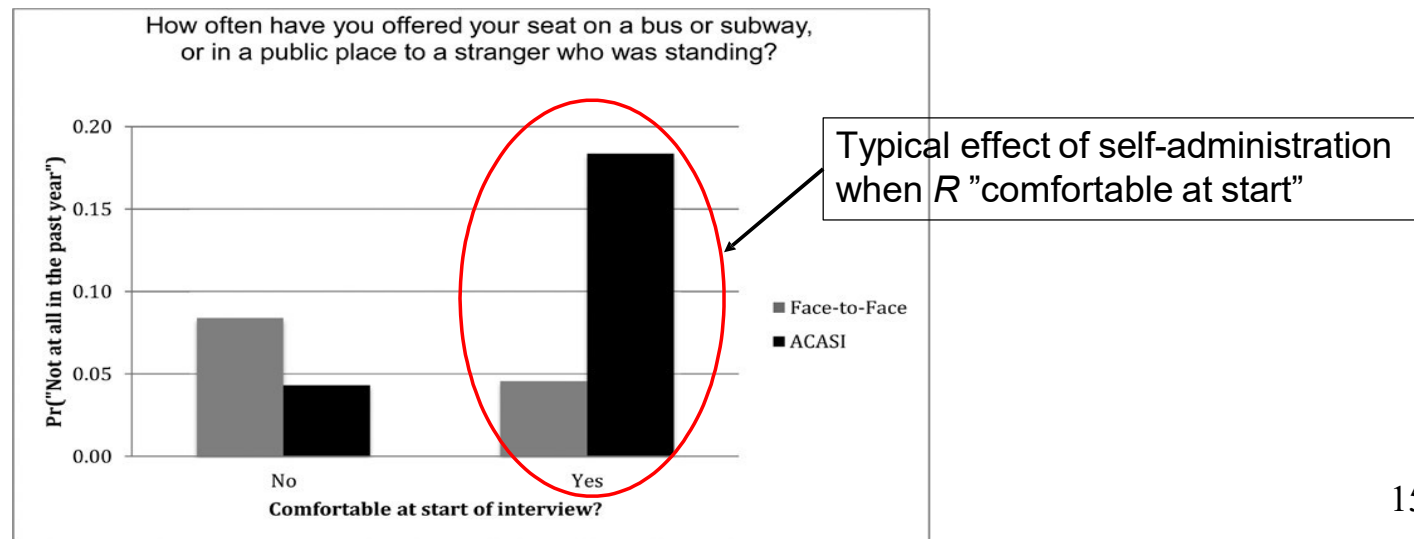
- ☐ Excellent
- ☐ Very good
- ☐ Good
- ☐ Fair
- ☐ Poor
- ☐ Don't know
- ☐ Not sure

Repeat Question

Next →

# Lind et al. (2013) Main Findings

- Most common pattern: less disclosure in modes w/ than w/o a face
  - *Rs* disclose less in 8 of 10 Qs in FTF and both *V*/s (HA and LA) than ACASI suggesting *V*/s create social presence
  - *Rs* disclose less in FTF than the three automated modes (HA, LA, ACASI) for only 4 of 10 questions (contradicts implicit prediction by Tourangeau & Smith)
- *Rs*' comfort affected impact of mode on disclosure



# Audio vs. Text:

## Does ACASI feel more private than CASI?

- Immersive audio (e.g., headphones) seems to disconnect listeners from surroundings
  - Cell phones and driving (Strayer & Johnston, 2001): cell phone conversations while driving led to 2-fold increase in failure to detect traffic signs and, when signals detected, slower RTs
- But Couper, Tourangeau & Marvin (2009) find no differences between ACASI and CASI (text-only) in reporting sensitive items in survey of fertility
  - May be due to non-use of audio and incomplete playing of audio in this study
  - No difference due to presence of others
  - No literacy advantage, at least those with lowest levels of education do not report differently in ACASI vs. text-CASI
  - So not clear that ACASI actually increases *Rs'* sense of privacy

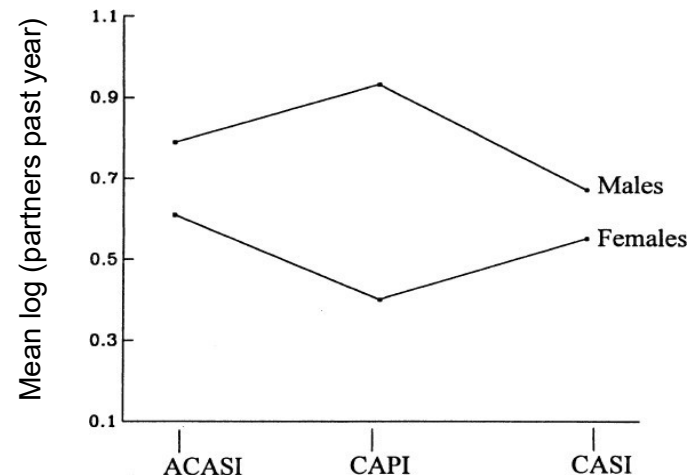


# Does ACASI increase honesty of answers or strategy used to answer sex partners question?

- Bogus pipeline study (T & S, 1998) suggests mode effects for sensitive Qs due to more and less honest reporting
  - Rs led to believe answers verified through polygraph-like machine
  - More sensitive behaviors reported in answering 15 of 19 Qs than in control
- But Brown and Sinclair (1999) suggest gender differences in reported number of sex partners due to gender-specific response strategies not social desirability
  - Males use “rough approximations” (general impressions) which lead to overestimation
    - Individual episodes seem hard for them to distinguish
  - Females enumerate partners which leads to underestimation
    - Individual episodes are likely distinctive and memorable

# Honesty vs. Response Strategy? (2)

- Possible both “motivated misreporting” (deception) and gender-specific strategies are involved
- Even under ACASI where social desirability should be low, some asymmetry remains



## Social Presence in Self administration?

- Does online self-administration (i.e., Web questionnaire) confer same sense of privacy as CASI and ACASI?
  - *Rs* could perceive disclosure on web to be very public and harmful in contrast to laptop in interview (i.e., CASI)
- Kreuter, Presser & Tourangeau (2009) found more honest reports (compared to records) in Web than IVR or CATI
  - asked student sample sensitive Qs, e.g., “At least one D or F?”
  - # “yes”: Web > IVR > CATI
  - “False negatives” based on records: Web < IVR < CATI

# Paradox of computerization

- CASI advantage presumably because computer alone lacks the social characteristics of a human intermediary
  - CASI and ACASI lack facial representation (Lind et al., 2013)
- but “Computers are social actors” (CASA) paradigm suggests that people ascribe human attributes to computers
  - e.g., Nass & Moon, 2000 report that people are (1) Polite in evaluating the computer’s performance, and (2) Ascribe gender stereotypes to computers based on voice
- How can this be reconciled?

# Ease of obtaining definitions

(Conrad, Couper, Tourangeau & Peytchev, 2006)

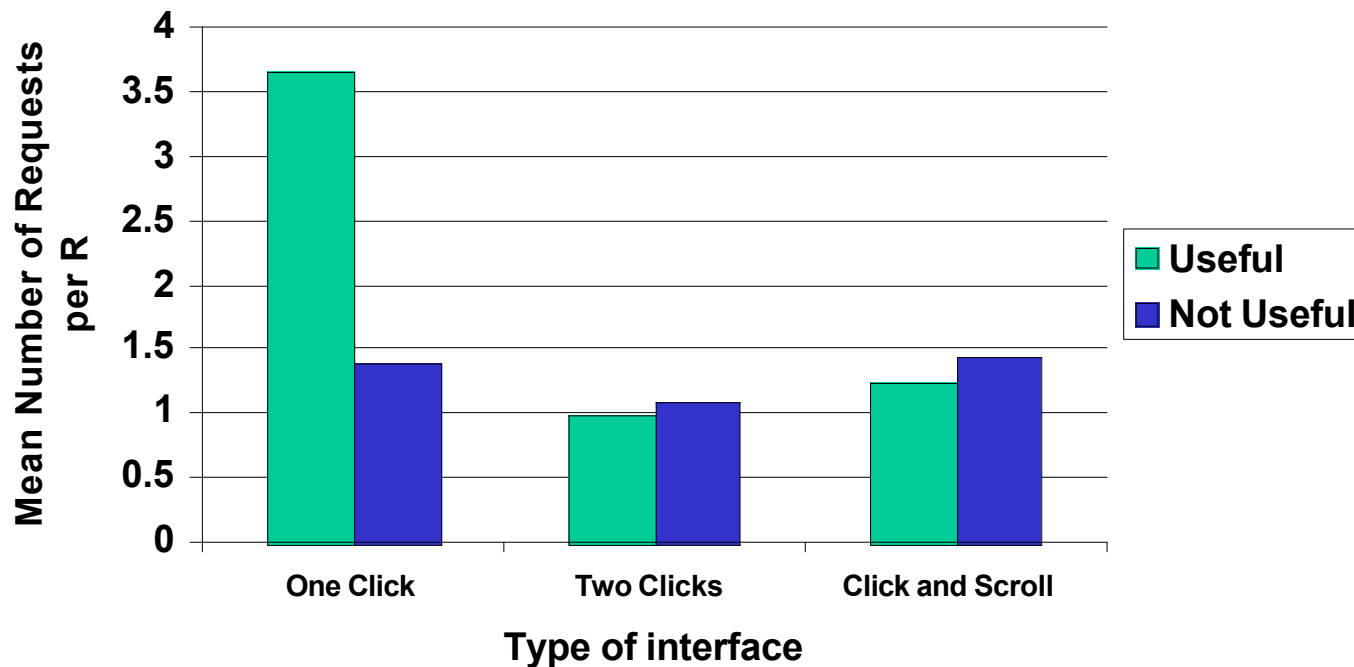
- Varied ease of getting definition
  - One click
  - Two clicks
  - Click and scroll
- Varied awareness of need for definition
  - Non-technical terms (e.g., beef, vegetables)
  - Technical terms (e.g., antioxidants, hydrogenated fats)
- Varied usefulness of definition
  - intuitive (“beef” includes food products from cows)
  - counterintuitive (“vegetables” includes French fries)

## Ease of obtaining definitions (2)

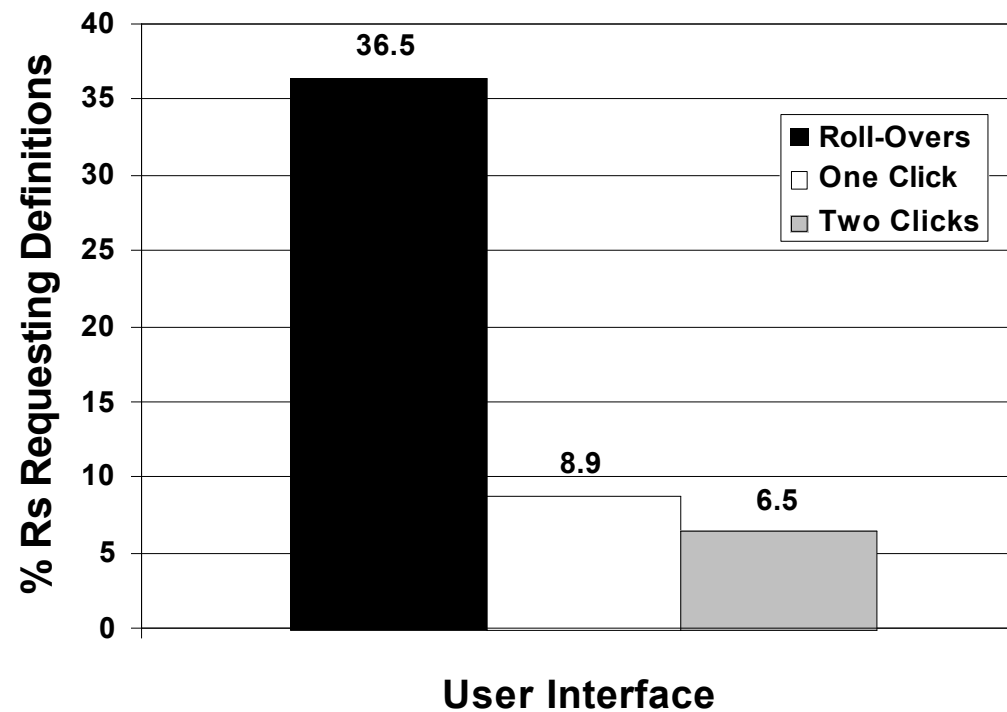
- Generally, few respondents make use of definitions
  - Only 17.4% of respondents accessed any definitions
- Use of definitions much higher for technical terms
  - 89% of definitions accessed for technical terms, 11% for non-technical
  - Presumably, technical terms increased awareness that definitions might be needed
- The fewer clicks required to get definition, the more *Rs* requested them
  - More *Rs* accessed (any) definitions when only one click required (56%) than two clicks (24%) or click plus scroll (20%)
  - *Rs* accessed more definitions when one click required (2.69) than two clicks (1.59) or click and scroll (1.47)

# Usefulness of definitions matters, if they are easy to obtain

## Ordinary Terms



Very low effort interface (mouse-overs) leads to more requests than even one click (exp. 2)

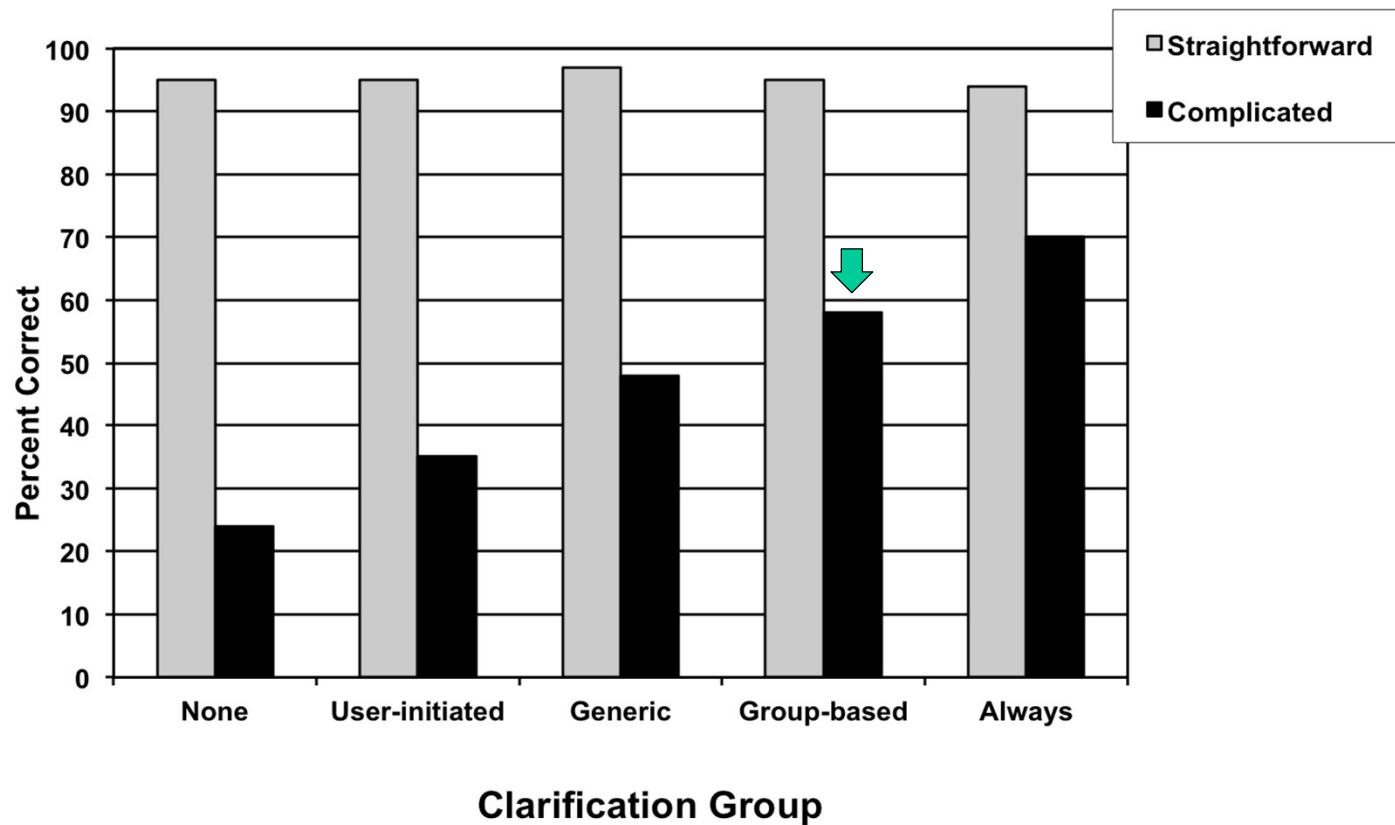




# System initiative and user modeling

- Can reduce effort/initiative required to obtain clarification by automatically presenting it, *if R seems uncertain*
- Conrad, Schober & Coiner (2007), exp. 2, operationalize *R*'s uncertainty as being *inactive* for longer than required to accurately answer *Q* on basis of a straightforward scenario
- Used a single inactivity threshold for all *Rs* in the “generic” treatment and different thresholds for faster (young) and slower (older) *Rs* in the “group-based” treatment
  - Shorter threshold for faster than slower group
- Prediction: clarification will help more when tailored for groups of *Rs*

# Initiative and User Modeling (2)

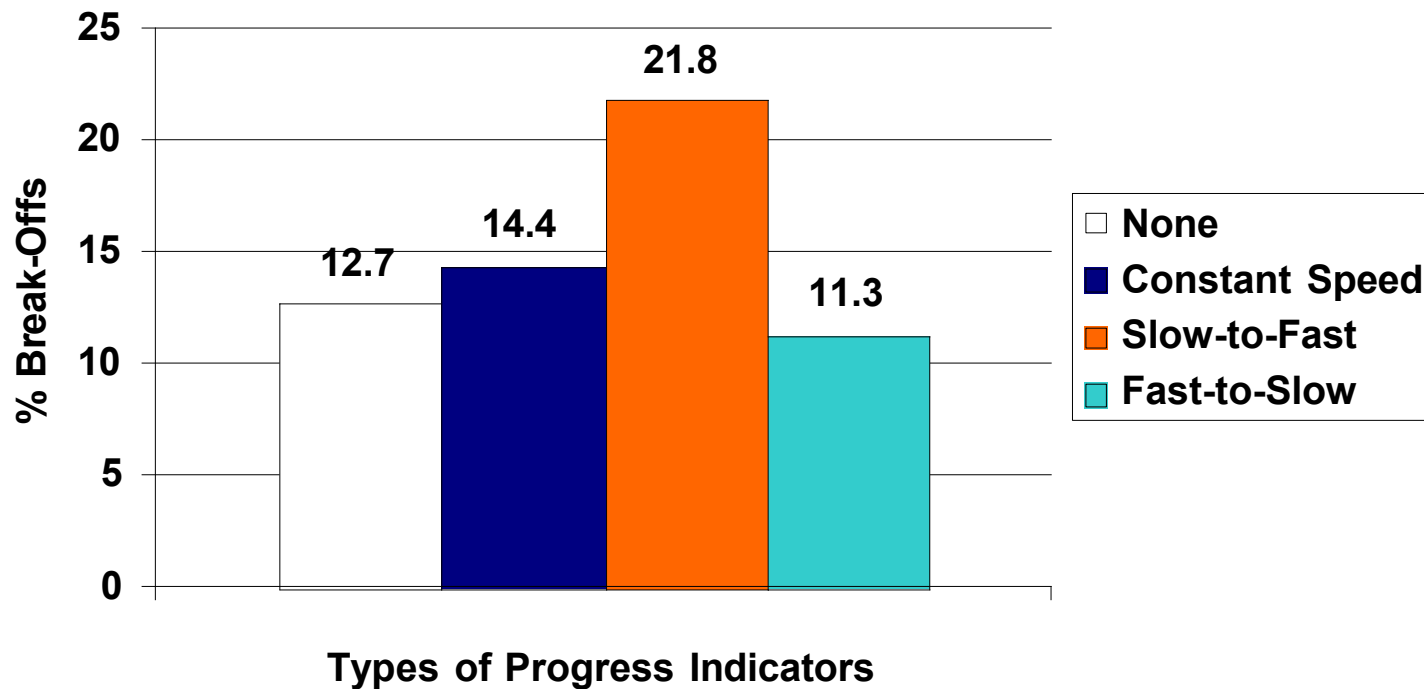


# Progress Indicators

- Couper, Traugott & Lamias (2000) proposed that progress indicators increase # completions, i.e., reduce abandonment
  - No significant difference in completion rates with or without progress indicators
  - Additional download time of graphic may have washed out advantage
  - Perhaps indication that  $R$  has a long way to go is not an incentive to complete

# Variable speed progress indicators

(Conrad, Couper, Tourangeau & Peytchev, 2010)



# Summary and Conclusions

- *I*wers seem to split their attention between computer and *R*
  - may reduce *I*wer's awareness of *R*'s experience
- Computerized self-administration seems to increase sense of privacy and thus honesty of answering sensitive questions
  - More undesirable answers do, in fact, reflect greater candor (compared to administrative records)
  - Increased disclosure seems to be more about absence of embodiment, especially a face, than automation/self-administration per se
- Interactive features like clickable definitions are used rarely by *Rs*
  - Even small differences in (already low) effort affect *Rs*' willingness to use interactive features
  - Progress Indicators seem to backfire, increasing breakoffs when they communicate that *R* still has a long way to go